

U.S. Airline Flight Performance & Delay Analytics

Q1 2026 • Exploratory Analytics + Predictive Modeling

Cameron J. Dillard

Data Analytics Portfolio Project



Project Scope

A wide analysis of U.S. domestic airline operations during Q1 2026.

1.85M

Q1 flight records

1.78M

completed flights modeled

21.59%

delayed 15+ minutes

4

classification models

Business questions

- How reliably are flights arriving on time?
- Which factors are associated or contribute to a delay?
- Can scheduled flight information predict a 15+ minute arrival delay?

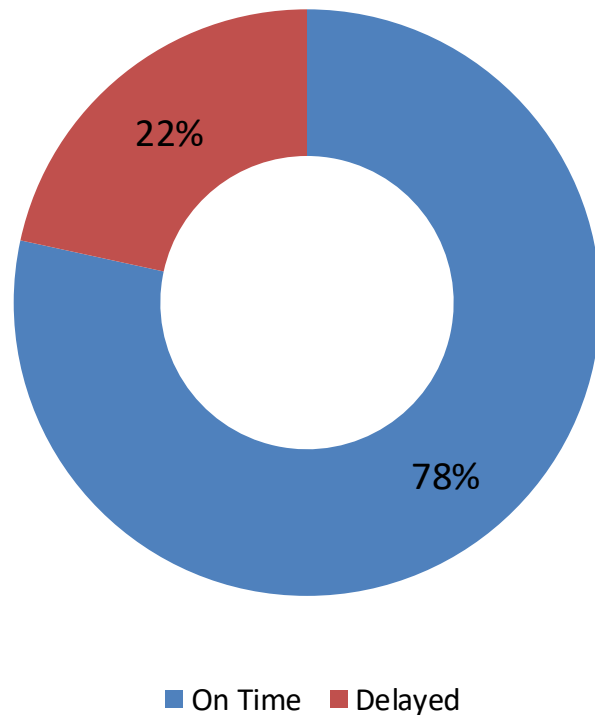
Predictors used

Month	Day of week
Marketing carrier	Origin
Destination	Departure hour
Distance	

Delay Classification Target

A flight is classified as delayed when arrival is 15 minutes or more after schedule.

Flights



Modeling dataset

78.41%

On time

21.59%

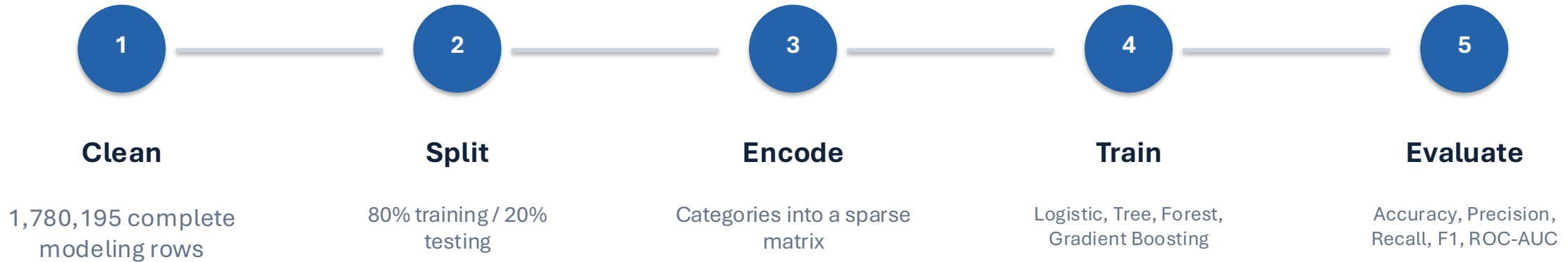
Delayed

The target is imbalanced, so accuracy alone is not sufficient. Recall, F1 score, and ROC-AUC are important for judging delay detection.

Train / test split: 80% / 20%
356,039 flights held out for testing

Predictive Modeling Workflow

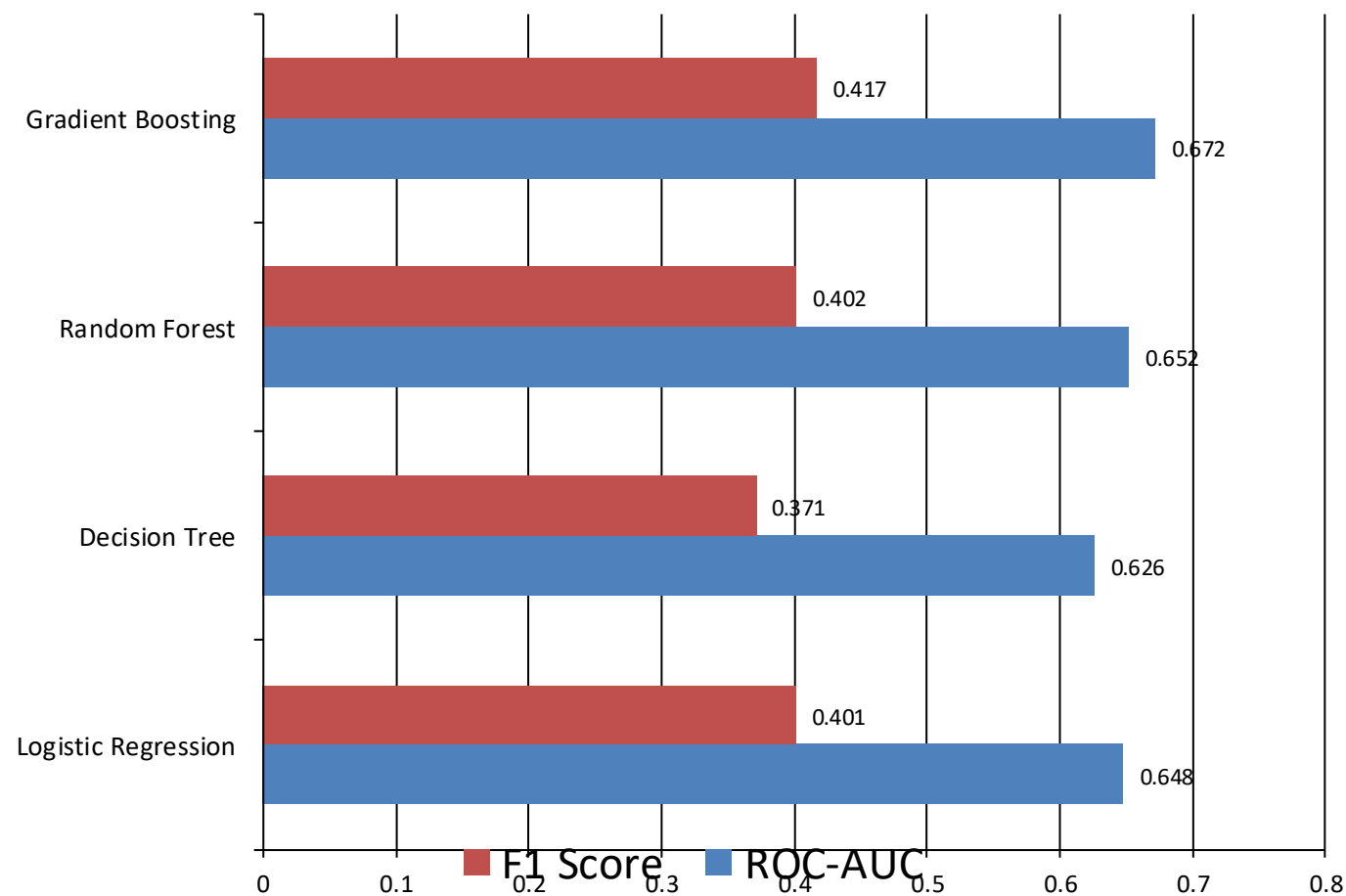
Seven cleaned predictors were transformed into 768 encoded model features.



Important design choice: only info available before departure was used, helping avoid leakage from actual arrival outcomes.

Model Performance Comparison

Gradient Boosting produced the strongest overall balance across the delay-focused metrics.



0.6722
Best ROC-AUC

0.4172
Best F1 score

Gradient Boosting also achieved 61.79% recall for delayed flights while improving precision to 31.49%.

Gradient Boosting: What the Predictions Mean

The model correctly identified 47,504 delayed flights in the test set.

175,789

Correct on-time

103,365

False delay alert

29,381

Missed delay

47,504

Correct delay

Interpretation

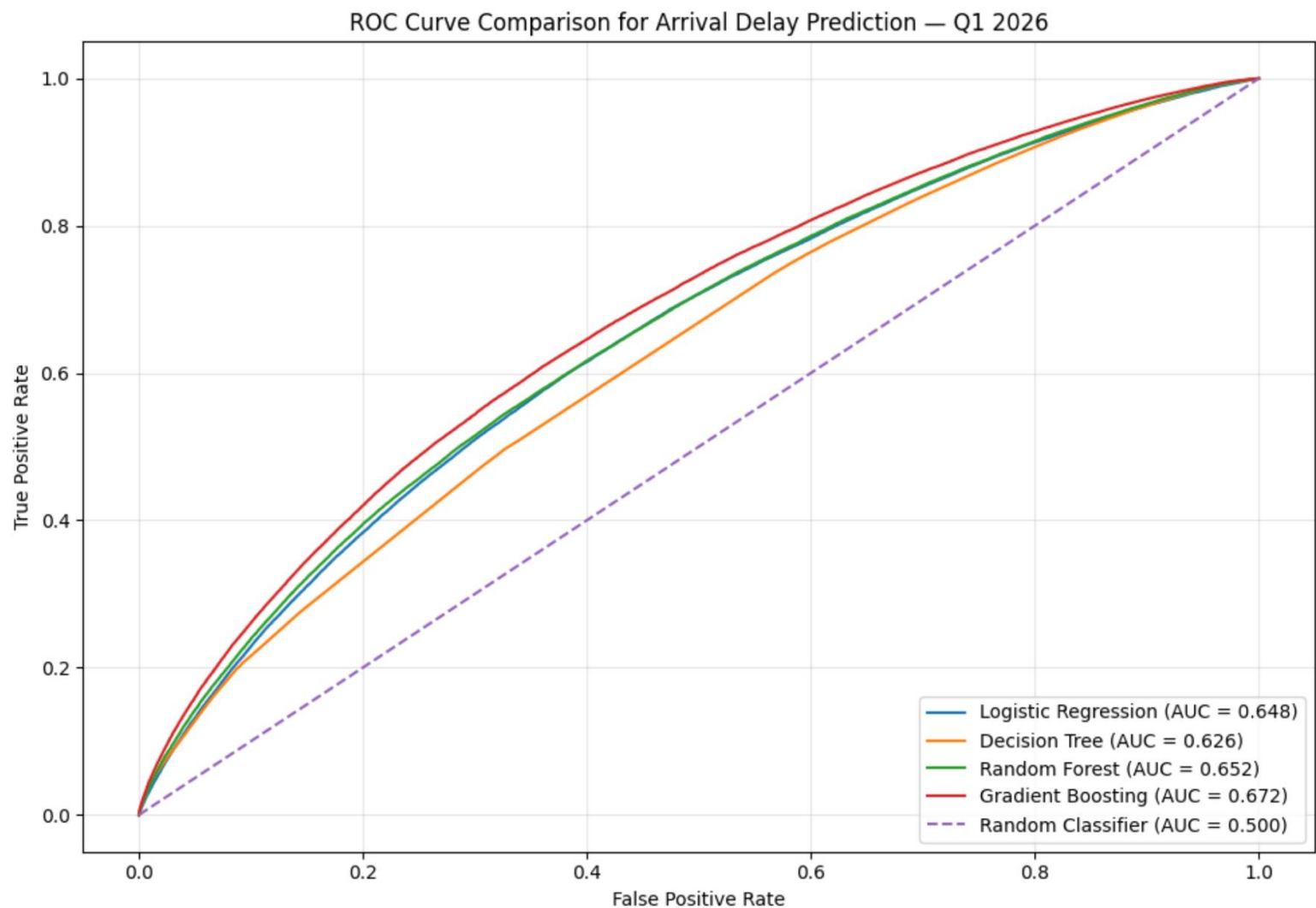
The model catches about 62% of delayed flights, but it also flags many on-time flights as delayed. This tradeoff is important when considering operational use.

Total correct predictions: 223,293

Total incorrect predictions: 132,746

ROC Analysis

All four models outperform random classification; Gradient Boosting separates the classes best.



0.672

Gradient Boosting AUC

0.652

Random Forest AUC

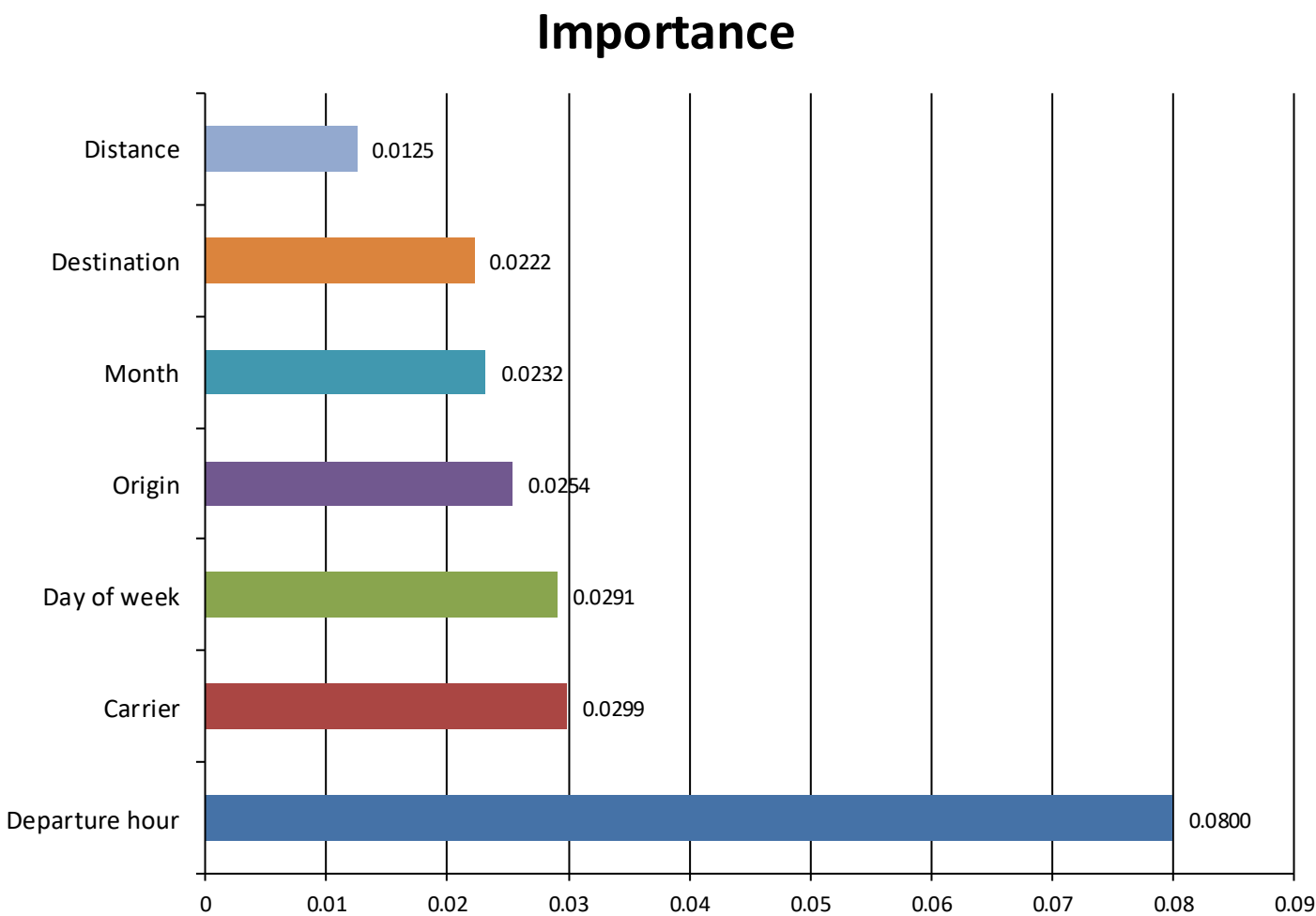
0.648

Logistic AUC

Decision Tree AUC: 0.626

What Drives Predicted Delay Risk?

Permutation importance shows departure hour as the strongest predictor in the final model.



Key insight

Departure hour has better predictive importance than any other single feature in the model.

Carrier, day of week, origin, month, and destination provide additional—but smaller—predictive value.

Key Takeaways

From 1.85 million Q1 records to an interpretable delay-risk model.

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|----|----------------------------|--|
| 01 | Delay is meaningful | 21.59% of completed flights arrived 15+ minutes late. |
| 02 | Best overall model | Gradient Boosting led ROC-AUC (0.6722) and F1 (0.4172). |
| 03 | Timing matters | Departure hour was the strongest feature-importance signal. |
| 04 | Room to improve | Future versions can add weather and historical route conditions. |

